

FPGA Implementation of Reconfigurable CORDIC Algorithm and a Memristive Chaotic System With Transcendental Nonlinearities

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Abstract—Coordinate Rotation Digital Computer (CORDIC) is a robust iterative algorithm that computes many transcendental mathematical functions. This paper proposes a reconfigurable CORDIC hardware design and FPGA realization that includes all possible configurations of the CORDIC algorithm. The proposed architecture is introduced in two approaches: multiplier-less and single multiplier approaches, each with its advantages. Compared to recent related works, the proposed implementation overpasses them in the included number of configurations. Additionally, it demonstrates efficient hardware utilization and suitability for potential applications. Furthermore, the proposed design is applied to a memristive chaotic system with different transcendental functions computed using the proposed reconfigurable block. The memristive system design is realized on the Artix-7 FPGA board, yielding throughputs of 0.4483 and 0.3972 Gbit/s for the two approaches of reconfigurable CORDIC.

Index Terms—Chaotic systems, CORDIC, FPGA, reconfigurable, transcendental.

I. INTRODUCTION

COORDINATE Rotation Digital Computer (CORDIC) is an iterative numerical method used to evaluate approximate values of transcendental mathematical functions. CORDIC reduces complex multiplication operations and, hence, it highly simplifies the overall hardware complexity [1], [2]. CORDIC applications include signal and image processing based on transforms such as: Fast Fourier Transform (FFT) [3], [4], Discrete Sine/Cosine Transforms (DST/DCT) [5]–[7] and Householder transform [8]. In addition, it is used in communication systems based on Orthogonal Frequency Division Multiplexing [9], [10] and real-time digital control systems. For example, intelligent robot control

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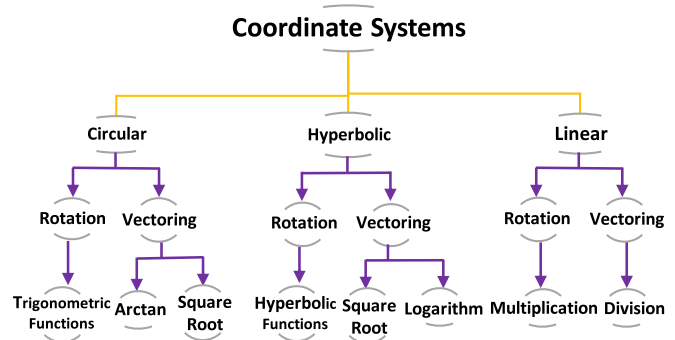


Fig. 1. Configurations of CORDIC algorithm.

systems require performing a large number of computations at high speed, where the robot must respond quickly to the environmental changes [11].

Different CORDIC configurations can calculate trigonometric functions, hyperbolic functions, square roots, multiplications, divisions, exponentials, and logarithms with arbitrary base, as presented in Fig. 1. CORDIC calculates the rotation and length of a vector r in linear, circular or hyperbolic coordinate systems, as illustrated in Fig. 2, depending on the function to be evaluated. Trajectories of the vector r_i for successive CORDIC iterations are shown in Fig. 2. CORDIC performs rotation iteratively using a series of specific incremental rotation angles selected so that each iteration is performed by shift and add operation. The vector norms preserving rotation trajectory in the circular, hyperbolic and linear coordinate systems are defined by the functions $x^2 + y^2 = 1$, $x^2 - y^2 = 1$ and $x = 1$, respectively. CORDIC configurations also include two different modes: the rotation mode and the vectoring mode [12]. The rotation mode is used to perform the general rotation by a given angle. The vectoring mode computes the unknown angle of a vector by performing a finite number of micro-rotations [12].

Several researches introduced Field-Programmable Gate Arrays (FPGA) implementations of the CORDIC algorithm with different enhancements. Some designs reduced the number of shifters and adders/subtractors by employing the Canonical Signed-Digit (CSD) technique and Hcub algorithm [2], Quantum-dot Cellular Automata (QCA) CORDIC [13], and dual-rotation iteration [14]. Additionally, others modified the adder/subtractor design to decrease the count of iterations and thus the hardware resources by using integrated

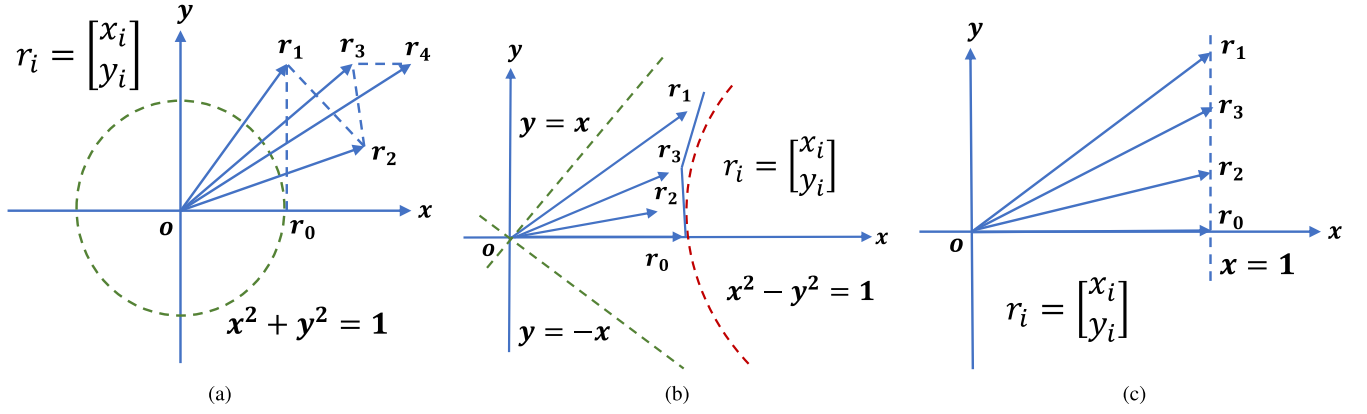


Fig. 2. Rotation of (a) Circular, (b) Hyperbolic and (c) Linear coordinate systems.

adder/subtractor in place of binary adder/subtractor [15] or by employing speculative adders for tactile data processing [16]. Furthermore, some researchers improved the throughput and reduced the latency of the design by using hybrid cascaded-recursive CORDIC [17] or pipelined CORDIC [18], [19].

FPGA realizations of digital transforms and applications employing CORDIC as a building block were presented including FFT [20], running DFT with refreshing mechanism [7], Discrete Fractional Fourier Transform (DFrFT) [21], Sliding Discrete Fourier Transform (SDFT) [22], DCT [23] and spike timing dependant plasticity (STDP) learning [24]. These researches focused on real-time computation with less power consumption, area overhead, and hardware complexity. In addition, CORDIC was employed in realizing rotation of chaotic attractors on FPGA [25].

These works [2], [7], [13]–[25] presented single-mode CORDIC architectures that work in a single coordinate system only. A different research direction targeted reconfigurability and proposed a dual-mode CORDIC including both circular and linear coordinates to be used in signal processing applications [26]. Other dual-mode CORDIC architectures included circular and hyperbolic coordinates [27], [28]. The authors in [29] introduced a unified reconfigurable CORDIC processor working in the three coordinates. However, it did not include the configuration responsible for computing hyperbolic functions, which are needed in various applications [27], [28], [30], specifically, chaotic systems with transcendental hyperbolic nonlinearities such as [31].

Lorenz first introduced the chaos phenomenon in 1960 [32]. Chaotic systems are sensitive to initial conditions such that any small variation eventually leads to a significant difference in behavior [33]. This characteristic is utilized in many fields such as communications systems [34], synchronization [35], encryption [36], [37] and traffic flow control [38]. With the rise of digital signal processors and FPGAs, chaotic systems realization in digital hardware has become more practical for industrial use [39], [40]. Fixed-point operations are widely used for hardware realizations to save costs and enhance speed [41].

This paper proposes the FPGA realization of a reconfigurable CORDIC that works in the three coordinate systems (circular-hyperbolic-linear). The reconfigurable design can be used in many applications, including chaotic systems,

communications, control, and others, requiring transcendental functions, such as trigonometric, hyperbolic, or division, using a simple and efficient method. The proposed reconfigurable CORDIC is implemented for 16, 25 and 32-bits and compared with recent related works. The 32-bit reconfigurable CORDIC design is implemented using the multiplier-less and single multiplier approaches. The proposed 32-bit reconfigurable CORDIC is applied to a memristive chaotic system proposed in [31]. It contains transcendental functions from the three coordinate systems. The whole design is experimentally validated and implemented on the Artix-7 FPGA board for the first time.

The rest of the paper is organized as follows: Section II introduces the general CORDIC algorithm and proposes the digital design and FPGA realization of the two approaches of reconfigurable CORDIC. Section III proposes the FPGA implementation of a memristive chaotic system based on reconfigurable CORDIC. Section IV shows the results and experimental setup, including the utilized resources, and compares them to recent related works. Finally, Section V summarizes the main contributions.

II. GENERAL CORDIC ALGORITHM AND PROPOSED RECONFIGURABLE ARCHITECTURE

General CORDIC algorithm is formulated as follows [42]:

$$x_{i+1} = k_n(x_i - \mu \sigma_i 2^{-i} y_i), \quad (1a)$$

$$y_{i+1} = k_n(y_i + \sigma_i 2^{-i} x_i), \quad (1b)$$

$$z_{i+1} = \begin{cases} z_i - \sigma_i \operatorname{arctanh}(2^{-i}), & \text{if } \mu = -1 \\ z_i - \sigma_i (2^{-i}), & \text{if } \mu = 0 \\ z_i - \sigma_i \operatorname{arctan}(2^{-i}), & \text{if } \mu = 1 \end{cases} \quad (1c)$$

where $\sigma_i = \operatorname{sign}(z_i)$ and $-\operatorname{sign}(y_i)$ for rotation and vectoring modes, respectively, i is the current iteration, n is the number of iterations, $\mu \in \{-1, 0, 1\}$ represents the hyperbolic, linear or circular coordinate system, respectively, and the scale factor k_n is calculated as follows [42]:

$$k_n = \prod_{i=0}^{n-1} \sqrt{1 + \mu 2^{-2i}}. \quad (2)$$

The computed scale factor for circular, hyperbolic and linear coordinates is $k_{n1} \simeq 1.64676$, $k_{n2} \simeq 0.82816$ and $k_{n3} = 1$,

TABLE I
DIFFERENT CONFIGURATIONS OF CORDIC ALGORITHM IN THE PROPOSED DESIGN

Operation	Configuration	SEL _{0,1,2}	Initialization	Assignment	Output	Post-processing
$\sin(a), \cos(a), \tan(a)$ $[-\pi : -\frac{\pi}{2}]$	C-RM	000	$x_0 = 0, y_0 = -\gamma_1$ $z_0 = a + \pi/2, m_1 = -1$ $m_2 = 1, m_3 = -1$	$m_0 = \text{sign}(z_i)$ $m_4 = \arctan(2^{-i})$ $m_5 = y_i$	$x_n = \cos(a)$ $y_n = \sin(a)$	$\tan(a) = \sin(a)/\cos(a)$
$\sin(a), \cos(a), \tan(a)$ $[-\frac{\pi}{2} : \frac{\pi}{2}]$	C-RM	001	$x_0 = \gamma_1, y_0 = 0$ $z_0 = a, m_1 = -1$ $m_2 = 1, m_3 = -1$	$m_0 = \text{sign}(z_i)$ $m_4 = \arctan(2^{-i})$ $m_5 = y_i$	$x_n = \cos(a)$ $y_n = \sin(a)$	$\tan(a) = \sin(a)/\cos(a)$
$\sin(a), \cos(a), \tan(a)$ $[\frac{\pi}{2} : \pi]$	C-RM	010	$x_0 = 0, y_0 = \gamma_1$ $z_0 = a - \pi/2, m_1 = -1$ $m_2 = 1, m_3 = -1$	$m_0 = \text{sign}(z_i)$ $m_4 = \arctan(2^{-i})$ $m_5 = y_i$	$x_n = \cos(a)$ $y_n = \sin(a)$	$\tan(a) = \sin(a)/\cos(a)$
$\sqrt{a^2 + 1}, \arctan(a)$	C-VM	011	$x_0 = \gamma_1, y_0 = a$ $z_0 = 0, m_1 = 1$ $m_2 = -1, m_3 = 1$	$m_0 = \text{sign}(y_i)$ $m_4 = \arctan(2^{-i})$ $m_5 = z_i$	$x_n = \sqrt{a^2 + 1}$ $z_n = \arctan(a)$	-
$\sinh(a), \cosh(a)$ $\tanh(a), \exp(a)$	H-RM	100	$x_0 = \gamma_2, y_0 = 0$ $z_0 = a, m_1 = 1$ $m_2 = 1, m_3 = -1$	$m_0 = \text{sign}(z_i)$ $m_4 = \text{arctanh}(2^{-i})$ $m_5 = y_i$	$x_n = \cosh(a)$ $y_n = \sinh(a)$	$\tanh(a) = \sinh(a)/\cosh(a)$ $\exp(a) = \cosh(a) + \sinh(a)$
$\frac{1}{2}\ln(a), 2\sqrt{a}$	H-VM	101	$x_0 = a + \gamma_2, y_0 = a - \gamma_2$ $z_0 = 0, m_1 = -1$ $m_2 = -1, m_3 = 1$	$m_0 = \text{sign}(y_i)$ $m_4 = \text{arctanh}(2^{-i})$ $m_5 = z_i$	$x_n = 2\sqrt{a}$ $z_n = \frac{1}{2}\ln(a)$	$\ln(a) = 2z_n$
ab	L-RM	110	$x_0 = a, y_0 = 0$ $z_0 = b, m_1 = 0$ $m_2 = 1, m_3 = -1$	$m_0 = \text{sign}(z_i)$ $m_4 = 2^{-i}$ $m_5 = y_i$	$z_n = ab$	-
b/a	L-VM	111	$x_0 = a, y_0 = b$ $z_0 = 0, m_1 = 0$ $m_2 = -1, m_3 = 1$	$m_0 = \text{sign}(y_i)$ $m_4 = 2^{-i}$ $m_5 = z_i$	$z_n = b/a$	-

respectively [42]. To perform scale factor compensation [43], let $\gamma_1 = 1/k_{n1}$ and $\gamma_2 = 1/k_{n2}$ for the coming initializations.

The proposed reconfigurable CORDIC architecture is based on the general CORDIC algorithm in (1) and achieves all possible configurations of the three coordinates in Fig. 1. All possible configurations are summarized in Table I. They are achieved through the variation of initial conditions and the values of μ , σ , z , and the signs in (1). Using scale factor compensation, the initialization of x_0 or y_0 in Table I is γ_1 or γ_2 for circular and hyperbolic coordinates, respectively, to reduce computation overhead in CORDIC algorithm [43]. For hyperbolic coordinates in vectoring mode, a is the input multiplied by γ_2 to balance the addition/subtraction in the initial conditions. The configurations in Table I include the circular coordinates in rotation mode that compute the trigonometric functions for input angle range from $-\pi$ to π , and in vectoring mode that computes the square root and inverse tangent functions. In addition, it includes the hyperbolic coordinates in rotation mode that compute the hyperbolic functions and in vectoring mode that computes Logarithmic and square root functions. Moreover, it includes the linear coordinates in rotation mode that compute the multiplication and vectoring mode that compute the division. The role of the division operation in Table I is highlighted in the case of the noninteger input, especially the post-processing operations that need fractional division to compute the tan and tanh functions in the circular and hyperbolic coordinates, respectively.

Figure 3 shows the proposed hardware design and architecture of a reconfigurable CORDIC algorithm. The design has two inputs, a clock, a reset signal, three selectors and two outputs. Three registers are used to store x , y , and z . The de-multiplexer with three selectors SEL_{0,1,2} chooses between different initializations, which leads to different configurations, and hence, outputs, (See Table I). SEL_{0,1,2} has two roles: enabling input range widening of the C-RM configuration in (1) from $[-\frac{\pi}{2} : \frac{\pi}{2}]$ to $[-\pi : \pi]$ using trigonometric

identities [44] and, furthermore, selecting the configuration among the six alternatives of Fig. 1. For C-RM in Table I, at the input range $a = [-\frac{\pi}{2} : \frac{\pi}{2}]$, the selected initial conditions are γ_1 , 0 and a . For the input range $a = [-\pi : -\frac{\pi}{2}]$, 0, $-\gamma_1$ and $a + \frac{\pi}{2}$ are chosen, while in the input range $a = [\frac{\pi}{2} : \pi]$, 0, γ_1 and $a - \frac{\pi}{2}$ are selected. The initialization and assignment of each function is stored in LUT as shown in the red dashed circle in Fig. 3. The value m_0 chooses between rotation and vectoring modes. The values m_1 , m_2 and m_3 represent the middle signs in x , y and z equations in (1). The value m_4 is the value of $\arctan(2^{-i})$, $\text{arctanh}(2^{-i})$ LUTs or shifting right to achieve the required function in Table I. The value m_5 determines the second output, whether to be y or z value according to Table I. In Fig. 3, the block denoted as F represents any multiplication operation in (1), where F can be implemented using two approaches. The multiplierless approach, denoted as approach A, uses one adder, two comparators, and two multiplexers for this block. The comparator selects the same input or its negation or zero. Approach A avoids using DSP multipliers and preserves the concept of using only shifters and adders/subtractors in the whole design. On the other hand, approach B uses one multiplier for the F block, allowing reconfigurable CORDIC to use fewer slices than approach A. The reconfigurable design uses three adder/subtractor blocks, three shifters, and three F blocks.

III. MEMRISTIVE CHAOTIC SYSTEM ARCHITECTURE

A memristive chaotic system presented in [31], is given by:

$$\dot{s} = u^2 - 0.5 v^2, \quad (3a)$$

$$\dot{u} = -v^2 - 1.4 u \tanh(w) + 4, \quad (3b)$$

$$\dot{v} = uv + 6.2 \cos(s), \quad (3c)$$

$$\dot{w} = u^2 - w. \quad (3d)$$

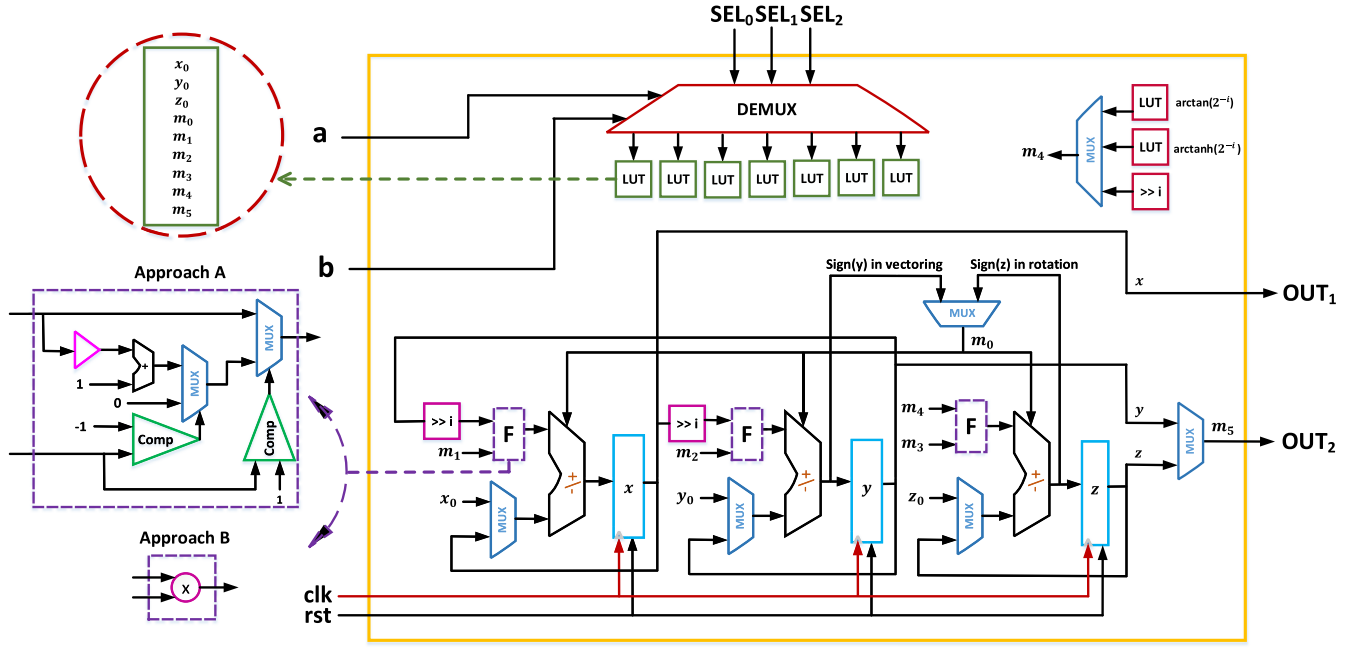


Fig. 3. Proposed hardware architecture of Reconfigurable CORDIC algorithm.

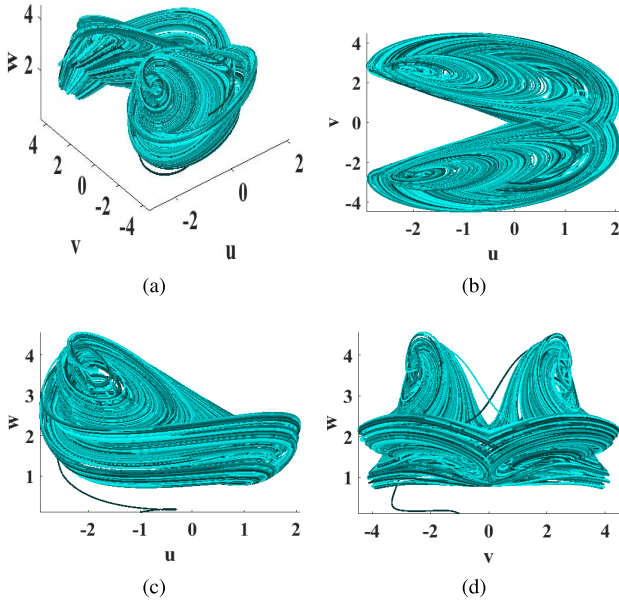


Fig. 4. Memristive system response (a) 3D view of uvw, (b) uv projection, (c) uw projection and (d) vw projection.

The system was not implemented in hardware before. It can be solved using Euler's numerical technique as follows:

$$s_{i+1} = s_i + h(u_i^2 - 0.5 v_i^2), \quad (4a)$$

$$u_{i+1} = u_i + h(-v_i^2 - 1.4 u_i \tanh(w_i) + 4), \quad (4b)$$

$$v_{i+1} = v_i + h(u_i v_i + 6.2 \cos(s_i)), \quad (4c)$$

$$w_{i+1} = w_i + h(u_i^2 - w_i). \quad (4d)$$

Figure 4 shows the memristive chaotic system response starting at initial conditions $(s_0, u_0, v_0, w_0) = (1 + \pi, -1, -1, 0)$ and $h = 0.01$.

As demonstrated in (4), the memristive system includes tanh function in (4b), which requires H-RM configuration for sinh and cosh computation and L-VM configuration for division operation to produce the tanh value. Additionally, the memristive system contains cos function in (4c), which needs C-RM configuration. Hence, the three coordinates are required for the memristive system hardware realization, easily implemented using the proposed reconfigurable CORDIC hardware block.

Figure 5 shows the proposed hardware architecture of the memristive system. Four registers are used to store s, u, v and w in 32-bit fixed point, 8b for the integer part and 24b for the fractional part. The proposed design requires five adders, four inverters, eight multipliers, and only one reconfigurable CORDIC block, which is required to perform the trigonometric and hyperbolic functions in (4). A select signal and control unit are used to maintain the reusability of the reconfigurable block in the same clk. In Fig. 5, the shape of the input arrows to the reconfigurable CORDIC block represents the type of the selected coordinate signals clarifying the switching between different configurations in the same clk. For the subtraction operation, two's complement is performed using an inverter and one as the adder's input carry.

IV. RESULTS AND DISCUSSION

Chaotic systems are highly sensitive to implementation aspects [45], [46]. Therefore, to specify the appropriate reconfigurable CORDIC data format for a sensitive chaotic system application, an error analysis is performed for the proposed reconfigurable CORDIC at 16, 25, and 32-bit fixed-point data format. The error measure is given by [47]:

$$\text{Relative Error} = \left| \frac{\text{ideal value} - \text{CORDIC value}}{\text{ideal value}} \right|. \quad (5)$$

The ideal value is calculated using MATLAB, and the CORDIC value is computed using the hardware simulation.

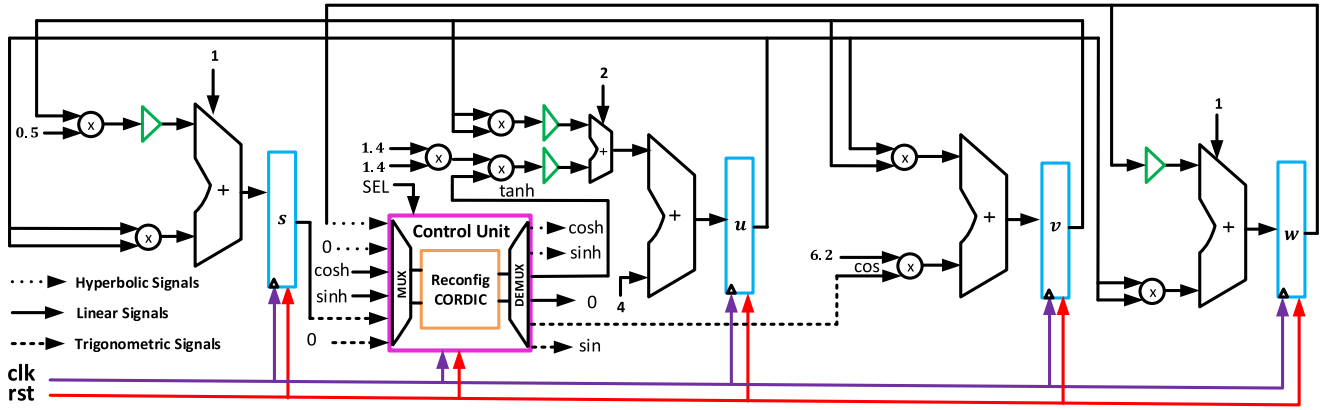


Fig. 5. Proposed hardware architecture of the memristive system.

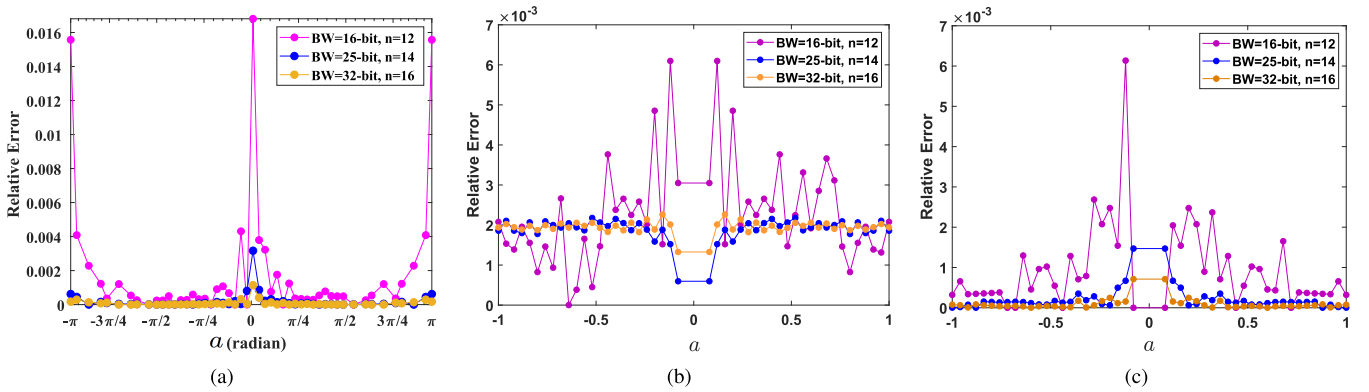


Fig. 6. Relative error of (a) sine, (b) sinh and (c) tanh using reconfigurable CORDIC.

The three cases of data format are validated for sine, sinh and tanh functions calculated by the proposed reconfigurable CORDIC. The test points taken for validation are 50 points equally spaced from $-\pi$ to π for sine and from -1 to 1 for sinh and tanh. The relative error graph for sine, sinh and tanh, respectively, is shown in Fig. 6. For bit-width (BW=16-bit) and $n = 12$, the maximum relative errors for the three functions $\approx 1.6\%$, 0.6% and 0.6% , respectively. For BW=25-bit and $n = 14$, the maximum relative errors $\approx 0.3\%$, 0.2% and 0.15% , respectively. For BW=32-bit and $n = 16$, the maximum relative errors $\approx 0.1\%$, 0.2% and 0.07% , respectively. Hence, the more bit-width, the less relative error, and the 32-fixed point format is selected for this application.

The proposed designs of reconfigurable CORDIC and the memristive system are written in Verilog HDL with the simulation of Xilinx ISE 14.7 and implemented on Xilinx FPGA Artix-7 XC7A100TCSG324 by using Chip scope. Figure 7 shows MATLAB floating-point software and Xilinx fixed-point hardware implementations of the memristive system. Both implementations preserve similar chaotic behavior in some points, while deviate slightly in other points due to hardware realization approximations.

Table II compares the FPGA resources of the proposed architecture using Approaches A, B to other recent reconfigurable CORDIC architectures for 16, 25, and 32-bit data format fixed point (for comparison purpose). The proposed reconfigurable design works in triple-mode, while the designs

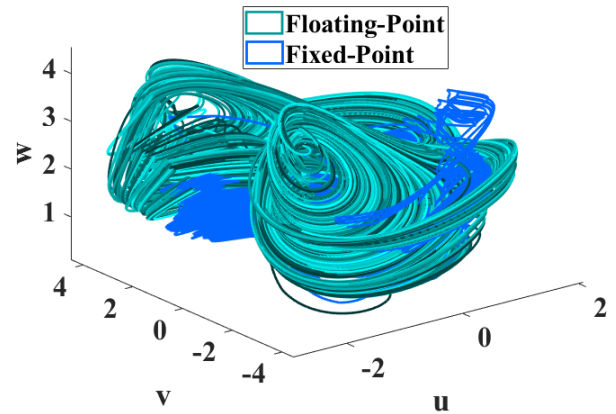


Fig. 7. MATLAB floating-point software and Xilinx fixed-point hardware implementations of the memristive system.

in [27], [28] work in dual-mode. While the proposed design covers all six configurations of Fig. 1, [29] covers only five configurations. In addition, the proposed reconfigurable architecture consumes fewer slice registers than the designs in [27], [29] and achieves a higher frequency than the one in [28], [29] for 16-bit format. Moreover, it consumes fewer slice registers and achieves a higher frequency (at approach A) than the one in [29] for 25-bit format. Comparing both approaches for the proposed architecture for the same bit-width, approach A achieves fewer slice LUTs, slice registers

TABLE II
COMPARISON OF RECONFIGURABLE CORDIC FPGA IMPLEMENTATIONS

Logic Utilization	Fixed-Point Data Format	FPGA Device	Configuration	Total No. Slices	Total No. Slice LUTs	Total No. Slice Regs	DSPs	Max Freq (MHz)	ARE
Aggarwal [27]	16-bits	Virtex-5 XC5VLX50-2FF153	C-RM / C-VM H-RM / H-VM	-	1460	898	-	323.76	10^{-4}
Aggarwal [28]	16-bits	Virtex-5 XC5VLX50-2FF153	C-RM / C-VM H-RM / H-VM	-	2161	369	-	142.14	10^{-3}
Fang [29]	16-bits	Virtex-5 XC5VFX130T	C-RM / C-VM L-RM / L-VM H-VM	-	1458	932	-	245.35	10^{-4}
This work (Approach A)	16-bits	Artix-7	C-RM / C-VM H-RM / H-VM	665	2202	645	0	276.510	10^{-3}
This work (Approach B)	16-bits	XC7A100TCSG324	L-RM / L-VM	710	2294	650	0	255.715	10^{-3}
Fang [29]	25-bits	Virtex-5 XC5VFX130T	C-RM / C-VM L-RM / L-VM H-VM	-	2169	1687	-	243.78	10^{-6}
This work (Approach A)	25-bits	Artix-7	C-RM / C-VM H-RM / H-VM	1320	4259	1154	0	255.563	10^{-4}
This work (Approach B)	25-bits	XC7A100TCSG324	L-RM / L-VM	1263	4303	1167	0	241.167	10^{-4}
This work (Approach A)	32-bits	Artix-7	C-RM / C-VM H-RM / H-VM	1710	5546	1570	0	245.44	10^{-5}
This work (Approach B)	32-bits	XC7A100TCSG324	L-RM / L-VM	1538	5615	1570	4	146.297	10^{-5}

TABLE III
FPGA RESOURCES SUMMARY OF THE MEMRISTIVE SYSTEM

Logic Utilization	Total No. of slices	Total No. of slice LUTs	Total No. of slice Registers	DSPs	Max Freq (MHz)	Throughput (Gbit/s)
Memristive system (A)	951	3440	374	84	14.009	0.4483
Memristive system (B)	725	2546	373	113	12.412	0.3972

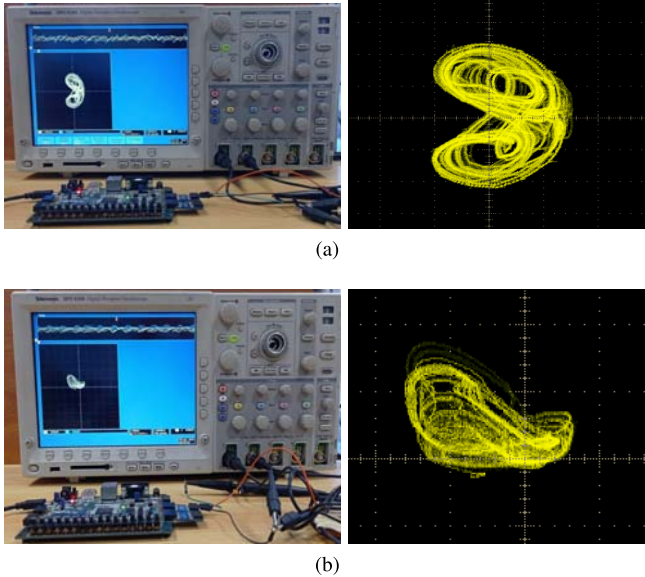


Fig. 8. FPGA experimental results of the memristive system (a) uv plane and (b) uw plane.

and higher frequency than B because the multiplier consumes more resources and time than the comparator. However, for 25-bits and 32-bits formats, approach B consumes fewer slices than A as the comparator consumes more logic than the multiplier. Generally, for approach B, the multiplier could consume resources from slices, slice LUTs and slice registers, as shown in 16-bits and 25-bits formats in Table II, or from DSPs, as shown in 23-bits format, which is decided by the

optimization process in Xilinx tool. The DSPs-consuming multiplier takes a longer time than the slices-consuming one for the 32-bits format with approach B. Table II also compares different recent works regarding the average relative error (ARE). In this work, ARE is computed using the average error values in the mentioned modes, which exhibits its maximum and minimum values at 16-bits and 32-bit cases, respectively..

Table III presents the FPGA resources of the memristive system implemented using the two approaches of the proposed reconfigurable CORDIC. While approach B consumes fewer slices, slice LUTs, and slice registers than A, the latter consumes fewer DSPs and achieves higher frequency and throughput. The achieved throughputs of the memristive system are 0.4483 and 0.3972 Gbit/s for approaches A and B, respectively. Figure 8 shows the experimental oscilloscope results of the memristive system using the Artix-7 FPGA board using approach A. The proposed reconfigurable CORDIC enables the reuse of the same block to compute different types of transcendental nonlinearities instead of multiple single-mode CORDIC blocks with different configurations.

V. CONCLUSION

The paper proposed the FPGA realization of a reconfigurable CORDIC achieving all possible configurations of the CORDIC algorithm for the three coordinates, shown in Fig. 1, in one generic design. The proposed implementation demonstrated efficient hardware utilization and suitability for potential applications compared to recent related works. Additionally, the proposed design was applied to a memristive chaotic system with different transcendental nonlinearities

that needed the integrity of functions computed using only the proposed triple-mode CORDIC design. The memristive system design was realized on the Artix-7 FPGA board, yielding throughputs of 0.4483 and 0.3972 Gbit/s for two design approaches of the reconfigurable CORDIC. Further reconfigurable CORDIC architecture enhancements and other fixed-point digital applications can be explored for future work.

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